1) a)

Points along the line with direction $\begin{pmatrix}1\\0\\1\end{pmatrix}$ are stretched by scale factor 2

1) b)

Points along the x-axis are stretched by scale factor 4

1) c)

The line with direction $\begin{pmatrix}1\\2\\-1\end{pmatrix}$ is invariant; This is a line of invariant points.

1) d)

The line with direction $\begin{pmatrix}0\\1\\-1\end{pmatrix}$ is invariant but points are reflected across the origin along the line.

1) e)

Points along the z-axis are stretched by scale factor 2 and points are reflected across the origin along the line.

2) a)

$$\begin{pmatrix}2&1&0\\0&3&0\\0&0&1\end{pmatrix}\begin{pmatrix}x\\y\\z\end{pmatrix}=\lambda\begin{pmatrix}x\\y\\z\end{pmatrix}$$

$$\mathbf{A}-\lambda\mathbf{I}=\begin{pmatrix}2&1&0\\0&3&0\\0&0&1\end{pmatrix}-\lambda\begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix}=\begin{pmatrix}2-\lambda&1&0\\0&3-\lambda&0\\0&0&1-\lambda\end{pmatrix}$$

$\det(\mathbf{A}-\lambda\mathbf{I})=(2-\lambda)(3-\lambda)(1-\lambda)$

So $(1-\lambda)(2-\lambda)(3-\lambda)=0$

$\therefore \lambda_1=1$, $\lambda_2=2$ or $\lambda_3=3$

When $\lambda_1=1$ then $\mathbf{A}-1\mathbf{I}=\begin{pmatrix}1&1&0\\0&2&0\\0&0&0\end{pmatrix}$

$$\begin{pmatrix}1&1&0\\0&2&0\\0&0&0\end{pmatrix}\begin{pmatrix}x\\y\\z\end{pmatrix}=0$$

$\Rightarrow x + y = 0$ (1)

$2y = 0$ (2)

$\therefore y = -x$ But since $y = 0$, $x = 0$

$\therefore$ when $\lambda_1 = 1$, $\mathbf{v}_1 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

When $\lambda_2 = 2$ then $\mathbf{A} - 2\mathbf{I} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$\Rightarrow y = 0$ (1)

$-z = 0$ (3)

$\therefore y = 0, z = 0$

$\therefore$ when $\lambda_2 = 2$, $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

When $\lambda_3 = 3$ then $\mathbf{A} - 3\mathbf{I} = \begin{pmatrix} -1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -2 \end{pmatrix}$

$$\begin{pmatrix} -1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$\Rightarrow -x + y = 0$ (1)

$-2z = 0$ (3)

$\therefore y = x, z = 0$

$\therefore$ when $\lambda_3 = 3$, $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$

$\therefore$ when $\lambda_1 = 1$, $\mathbf{v}_1 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$, when $\lambda_2 = 2$, $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, when $\lambda_3 = 3$, $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$

2) b)

$$\det \mathbf{A} = \begin{vmatrix} 2 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 2 \times 3 \times 1 = 6$$

The transformation scales the volume by a factor of 6

2) c)

The eigenvalues and eigenvectors tell us:

- Points along the line with direction $\begin{pmatrix}1\\0\\0\end{pmatrix}$ are stretched by scale factor 2
- Points along the line with direction $\begin{pmatrix}1\\1\\0\end{pmatrix}$ are stretched by scale factor 3
- The z-axis is invariant; This is a line of invariant points.

3) a)

$$\begin{pmatrix}-1 & 2 & -1\\ 0 & 1 & 1\\ 0 & 0 & 2\end{pmatrix}\begin{pmatrix}x\\y\\z\end{pmatrix}=\lambda\begin{pmatrix}x\\y\\z\end{pmatrix}$$

$$\mathbf{A}-\lambda\mathbf{I}=\begin{pmatrix}-1 & 2 & -1\\ 0 & 1 & 1\\ 0 & 0 & 2\end{pmatrix}-\lambda\begin{pmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}=\begin{pmatrix}-1-\lambda & 2 & -1\\ 0 & 1-\lambda & 1\\ 0 & 0 & 2-\lambda\end{pmatrix}$$

$\det(\mathbf{A}-\lambda\mathbf{I})=(-1-\lambda)(1-\lambda)(2-\lambda)$

So $(-1-\lambda)(1-\lambda)(2-\lambda)=0$

$\therefore \lambda_1=-1,\ \lambda_2=1$ or $\lambda_3=2$

When $\lambda_1=-1$ then $\mathbf{A}+1\mathbf{I}=\begin{pmatrix}0 & 2 & -1\\ 0 & 2 & 1\\ 0 & 0 & 3\end{pmatrix}$

$$\begin{pmatrix}0 & 2 & -1\\ 0 & 2 & 1\\ 0 & 0 & 3\end{pmatrix}\begin{pmatrix}x\\y\\z\end{pmatrix}=0$$

$\Rightarrow 2y-z=0$ (1)

$2y+z=0$ (2)

$3z=0$ (3)

$\therefore z=0\Rightarrow y=0$

$\therefore$ when $\lambda_1=1,\ \mathbf{v}_1=\begin{pmatrix}1\\0\\0\end{pmatrix}$

When $\lambda_2=1$ then $\mathbf{A}-1\mathbf{I}=\begin{pmatrix}-2 & 2 & -1\\ 0 & 0 & 1\\ 0 & 0 & 1\end{pmatrix}$

$$\begin{pmatrix}-2 & 2 & -1\\ 0 & 0 & 1\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}x\\y\\z\end{pmatrix}=0$$

$\Rightarrow -2x+2y-z=0$ (1)

$z = 0$ (2)

$\therefore x = y, z = 0$

$\therefore$ when $\lambda_2 = 1$, $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$

When $\lambda_3 = 2$ then $\mathbf{A} - 2\mathbf{I} = \begin{pmatrix} -3 & 2 & -1 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$

$\begin{pmatrix} -3 & 2 & -1 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$

$\Rightarrow -3x + 2y - z = 0$ (1)

$-y + z = 0$ (2)

$\therefore z = y$

Sub into (1): $-3x + y = 0 \therefore y = 3x$

$\therefore$ when $\lambda_3 = 2$, $\mathbf{v}_3 = \begin{pmatrix} 1 \\ 3 \\ 3 \end{pmatrix}$

$\therefore$ when $\lambda_1 = -1$, $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, when $\lambda_2 = 1$, $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$, when $\lambda_3 = 2$, $\mathbf{v}_3 = \begin{pmatrix} 1 \\ 3 \\ 3 \end{pmatrix}$

3) b)

$\det \mathbf{A} = \begin{vmatrix} -1 & 2 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{vmatrix} = -1 \times 1 \times 2 = -2$

The transformation scales the volume by a factor of 2 and orientation is reversed.

3) c)

The eigenvalues and eigenvectors tell us:

- Points along the x-axis are reflected across the origin along the axis.
- The line with direction $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ is invariant; This is a line of invariant points.
- Points along the line with direction $\begin{pmatrix} 1 \\ 3 \\ 3 \end{pmatrix}$ are stretched by scale factor 2

4) a)

$$\begin{pmatrix}3 & -4 & -2\\ 0 & -1 & 0\\ 0 & 2 & 1\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix}=\lambda\begin{pmatrix}x\\ y\\ z\end{pmatrix}$$

$$\mathbf{A}-\lambda\mathbf{I}=\begin{pmatrix}3 & -4 & -2\\ 0 & -1 & 0\\ 0 & 2 & 1\end{pmatrix}-\lambda\begin{pmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}=\begin{pmatrix}3-\lambda & -4 & -2\\ 0 & -1-\lambda & 0\\ 0 & 2 & 1-\lambda\end{pmatrix}$$

$\det(\mathbf{A}-\lambda\mathbf{I})=(3-\lambda)(-1-\lambda)(1-\lambda)$

So $(3-\lambda)(-1-\lambda)(1-\lambda)=0$

$\therefore \lambda_1=-1,\ \lambda_2=1$ or $\lambda_3=3$

When $\lambda_1=-1$ then $\mathbf{A}+1\mathbf{I}=\begin{pmatrix}4 & -4 & -2\\ 0 & 0 & 0\\ 0 & 2 & 2\end{pmatrix}$

$$\begin{pmatrix}4 & -4 & -2\\ 0 & 0 & 0\\ 0 & 2 & 2\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix}=0$$

$\Rightarrow 4x-4y-2z=0$ (1)

$2y+2z=0$ (3)

$\therefore y=-z$

Sub into (1): $4x+2z=0\Rightarrow z=-2x$

$\therefore$ when $\lambda_1=-1$, $\mathbf{v}_1=\begin{pmatrix}1\\ 2\\ -2\end{pmatrix}$

When $\lambda_2=1$ then $\mathbf{A}-1\mathbf{I}=\begin{pmatrix}2 & -4 & -2\\ 0 & -2 & 0\\ 0 & 2 & 0\end{pmatrix}$

$$\begin{pmatrix}2 & -4 & -2\\ 0 & -2 & 0\\ 0 & 2 & 0\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix}=0$$

$\Rightarrow 2x-4y-2z=0$ (1)

$-2y=0$ (2)

$\therefore y=0,\ x=z$

$\therefore$ when $\lambda_2=1$, $\mathbf{v}_2=\begin{pmatrix}1\\ 0\\ 1\end{pmatrix}$

When $\lambda_3=3$ then $\mathbf{A}-3\mathbf{I}=\begin{pmatrix}0 & -4 & -2\\ 0 & -4 & 0\\ 0 & 2 & -2\end{pmatrix}$

$$\begin{pmatrix}0 & -4 & -2\\ 0 & -4 & 0\\ 0 & 2 & -2\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix}=0$$

$\Rightarrow -4y - 2z = 0$ (1)

$-4y = 0$ (2)

$2y - 2z = 0$ (3)

$\therefore y = 0$ and $z = y$ so $z = 0$

$\therefore$ when $\lambda_3 = 3$, $\mathbf{v}_3 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

$\therefore$ when $\lambda_1 = -1$, $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$, when $\lambda_2 = 1$, $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$, when $\lambda_3 = 3$, $\mathbf{v}_3 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

4) b)

$\det \mathbf{A} = \begin{vmatrix} 3 & -4 & -2 \\ 0 & -1 & 0 \\ 0 & 2 & 1 \end{vmatrix} = 3 \times -1 \times 1 = -3$

The transformation scales the volume by a factor of 3 and orientation is reversed.

4) c)

The eigenvalues and eigenvectors tell us:

- The line with direction $\begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$ is invariant but points are reflected across the origin along the line.
- The line with direction $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ is invariant; This is a line of invariant points.
- Points along the x-axis are stretched by scale factor 3

5) a)

Since $-5\begin{pmatrix} 3 \\ 4 \\ -1 \end{pmatrix} = \begin{pmatrix} -15 \\ -20 \\ 5 \end{pmatrix}$ then $\begin{pmatrix} 3 \\ 4 \\ -1 \end{pmatrix}$ is an eigenvector.

5) b)

$\lambda = -5$

6) a)

- The line with direction $\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$ are stretched by scale factor 2^n
- The x-axis is invariant; points on this line remain unchanged.
- Points along the z-axis are stretched by scale factor $(-3)^n$
  (3^n with a reflection if n is odd)

6) b)

$\mathbf{P} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix}$, $\mathbf{D} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -3 \end{pmatrix}$ and $\mathbf{P}^{-1} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}$

$\mathbf{A} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix}\begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -3 \end{pmatrix}\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix}\begin{pmatrix} 0 & 2 & 0 \\ 1 & 0 & 0 \\ 0 & -3 & -3 \end{pmatrix} =$

$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & -5 & -3 \end{pmatrix}$

$\therefore \mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & -5 & -3 \end{pmatrix}$

6) c)

$\mathbf{A}^n = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix}\begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -3 \end{pmatrix}^n \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix} =$

$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix}\begin{pmatrix} 2^n & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & (-3)^n \end{pmatrix}\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}$

$= \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix}\begin{pmatrix} 0 & 2^n & 0 \\ 1 & 0 & 0 \\ 0 & (-3)^n & (-3)^n \end{pmatrix}$

$\therefore \mathbf{A}^n = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2^n & 0 \\ 0 & -2^n + (-3)^n & (-3)^n \end{pmatrix}$

7) a)

$$\begin{pmatrix} p & q & 0 \\ q & p & 0 \\ 0 & 0 & r \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda\begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} p & q & 0 \\ q & p & 0 \\ 0 & 0 & r \end{pmatrix} - \lambda\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} p-\lambda & q & 0 \\ q & p-\lambda & 0 \\ 0 & 0 & r-\lambda \end{pmatrix}$$

$\det(\mathbf{A} - \lambda\mathbf{I}) = (p-\lambda)(p-\lambda)(r-\lambda) - q(q)(r-\lambda)$

$= (r-\lambda)\big((p-\lambda)(p-\lambda) - q(q)\big)$

$= (r-\lambda)(p^2 - 2p\lambda + \lambda^2 - q^2)$

$= (r-\lambda)\big(\lambda - (p+q)\big)\big(\lambda - (p-q)\big)$

So $(r-\lambda)\big(\lambda - (p+q)\big)\big(\lambda - (p-q)\big) = 0$

$\therefore \lambda_1 = p-q$, $\lambda_2 = p+q$ or $\lambda_3 = r$

When $\lambda_1 = p-q$ then $\mathbf{A} - (p-q)\mathbf{I} = \begin{pmatrix} q & q & 0 \\ q & q & 0 \\ 0 & 0 & r-p+q \end{pmatrix}$

$$\begin{pmatrix} q & q & 0 \\ q & q & 0 \\ 0 & 0 & r-p+q \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$\Rightarrow qx + qy = 0$ (1)

$(r-p+q)z = 0$ (3)

$\therefore y = -x$ and $z = 0$

$\therefore$ when $\lambda_1 = p-q$, $\mathbf{v}_1 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$

When $\lambda_2 = p+q$ then $\mathbf{A} - (p+q)\mathbf{I} = \begin{pmatrix} -q & q & 0 \\ q & -q & 0 \\ 0 & 0 & r-p-q \end{pmatrix}$

$$\begin{pmatrix} -q & q & 0 \\ q & -q & 0 \\ 0 & 0 & r-p-q \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$\Rightarrow -qx + qy = 0$ (1)

$qx - qy = 0$ (2)

$(r-p-q)z = 0$ (3)

$\therefore z = 0$, $y = x$

$\therefore$ when $\lambda_2 = p+q$, $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$

When $\lambda_3 = r$ then $\mathbf{A} - r\mathbf{I} = \begin{pmatrix} p-r & q & 0 \\ q & p-r & 0 \\ 0 & 0 & 0 \end{pmatrix}$

$\begin{pmatrix} p-r & q & 0 \\ q & p-r & 0 \\ 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$

$\Rightarrow (p-r)x + qy = 0 \quad (1)$

$qx + (p-r)y = 0 \quad (2)$

From (2): $y = \frac{q}{r-p}x$

Sub into (1): $(p-r)x + q\left(\frac{q}{r-p}x\right) = 0$

so for distinct eigenvalues ($r \neq p \pm q$), $x = y = 0$

$\therefore$ when $\lambda_3 = r$, $\mathbf{v}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

$\therefore$ when $\lambda_1 = p - q$, $\mathbf{v}_1 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$, when $\lambda_2 = p + q$, $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$, when $\lambda_3 = r$, $\mathbf{v}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

7) b)

$\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = 1 - 1 + 0 = 0$ so $\mathbf{v}_1$ and $\mathbf{v}_2$ are perpendicular.

$\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = 0 - 0 + 0 = 0$ so $\mathbf{v}_1$ and $\mathbf{v}_3$ are perpendicular.

$\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = 0 + 0 + 0 = 0$ so $\mathbf{v}_2$ and $\mathbf{v}_3$ are perpendicular.

7) c)

The eigenvalues and eigenvectors tell us:

- Points along the line with direction $\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$ are stretched by scale factor $p - q$
- Points along the line with direction $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ are stretched by scale factor $p + q$
- Points along the z-axis are stretched by scale factor r

8) a)

$$\begin{pmatrix} 1-k & k & k \\ -2 & k+1 & 2 \\ 2-2k & k & 2k-1 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda\begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} 1-k & k & k \\ -2 & k+1 & 2 \\ 2-2k & k & 2k-1 \end{pmatrix} - \lambda\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1-k-\lambda & k & k \\ -2 & k+1-\lambda & 2 \\ 2-2k & k & 2k-1-\lambda \end{pmatrix}$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = \begin{vmatrix} 1-k-\lambda & k & k \\ -2 & k+1-\lambda & 2 \\ 2-2k & k & 2k-1-\lambda \end{vmatrix}$$

$$= \begin{vmatrix} 1-k-\lambda & k & k \\ -2 & k+1-\lambda & 2 \\ 1-k+\lambda & 0 & k-1-\lambda \end{vmatrix}$$

$$= \begin{vmatrix} 1-k-\lambda & k & 1-\lambda \\ -2 & k+1-\lambda & 0 \\ 1-k+\lambda & 0 & 0 \end{vmatrix}$$

$$= 0 - 0 + (1-\lambda)\big(-(k+1-\lambda)(1-k+\lambda)\big)$$

$$= (1-\lambda)(k+1-\lambda)(k-1-\lambda)$$

$$\therefore \lambda_1 = 1,\ \lambda_2 = k-1 \text{ or } \lambda_3 = k+1$$

8) b)

The eigenvalues and eigenvectors tell us:

- The line with direction $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ is invariant; This is a line of invariant points.
- Points along the line with direction $\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$ are stretched by scale factor $(k-1)^3$
- Points along the line with direction $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ are stretched by scale factor $(k+1)^3$